PhytoPi CLC - Functional and Nonfunctional Requirements

Nolan Tuttle & Daniel Grijalva

SWE-310: Software Engineering 1

Professor Hughes

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Jira Account

Jira Account: https://plantgrowthmanager.atlassian.net/jira/software/projects/D4N/summary

Roles and Responsibilities

Team Role	Name	Signature	Date
Hardware/ Integration Lead	Daniel Grijalva	Daniel Grijalva	10/05/25
Software/ Cloud Lead	Nolan Tuttle	Nolan Tuttle	10/05/25

Functional and Non-Functional Requirements

Requirement	Key	Acceptance Criteria	Type of Requirement	Input Requirements	Assumptions	Priority
The system shall provide a complete electrical wiring diagram for all hardware components.	D4N-229	AC: Given the list of hardware components, a complete wiring diagram is created covering all connections.	FR	Electrical schematics, component datasheets, hardware manuals	All hardware components are available and correctly specified	Medium
The system shall highlight all important connections such as signal and power so that the diagram is properly organized.	D4N-230	AC: Safety warnings are displayed for all sensitive or high-voltage components.	FR	Electrical schematics, signal and power connection notes, wiring diagrams	Electrical connections are standard and documented	Medium
The system shall indicate safety measures for components prone to damage or with high voltage.	D4N-231	AC: The created diagram is organized with labels and clear to read.	FR	Component datasheets, high-voltage warnings, safety manuals	High voltage components are correctly rated and protected	Highest
The system shall read temperature from temperature	D4N-233	AC: Temperature readings are captured every 5 minutes.	FR	Temperature sensors, sensor IDs, timestamps,	Temperature sensors are installed	High

sensors at specified intervals.				data acquisition interface	and functional	
The system shall generate recommendations for adjustment based on analysis results.	D4 N-332	AC: Recommendations are generated for all plants with ,â§3-second processing time.	FR	Historical sensor readings, analysis software, computation outputs	Historical plant sensor and image data is accurate and complete	Medium
The system shall analyze historical pH data to determine imbalances.	D4 N-331	AC: Analysis results correctly identify out-of-range pH levels.	FR	Historical pH data, sensor IDs, timestamps, calibration reference values	Historical pH data is available and reliable	Medium
The system shall train a machine learning model using historical plant sensor and image data to predict growth and stress.	D4 N-329	AC: Model successfully trains on available data with no runtime errors.	FR	Historical plant sensor readings, labeled image datasets, ML training software	Training datasets are labeled and representative of plant growth conditions	High
The system shall require authentication tokens for each data access requests.	D4 N-326	AC: Unauthorized requests return the proper HTTPS error codes.	FR	User authentication tokens, system access logs, credentials database	Users are authenticated and access tokens are valid	Highest
The system shall provide secure API endpoints for retrieving sensor data, alerts, or ML predictions.	D4 N-325	AC: API calls can be made to retrieve data successfully and requests are authenticated before returning data.	FR	API request structures, authentication tokens, secure network configuration	API endpoints are accessible and secure	Highest
The system shall store all machine learning predictions in the cloud database.	D4 N-323	AC: Predictions are successfully written to the database with correct metadata.	FR	Cloud database access, authentication	Cloud storage is available and	High

				credentials, ML prediction data	properly configured	
The system shall adjust artificial grow light to automatically maintain lighting target values.	D4 N-282	AC: The system automatically adjusts so that target light levels remain within 5% of the target value.	FR	Grow light controller interface, target lighting values, sensor feedback	Grow lights are installed and controllable	High
The system shall compare received light sensor readings to target light levels.	D4 N-281	AC: The system successfully detects when light levels are out of range within 2 seconds of deviation.	FR	Light sensor readings, target lighting values, timestamps	Light sensors are installed and calibrated	Medium
The system shall record nutrient-related alerts/anomalies for trend analysis	D4 N-278	AC: Alert logs for pH levels are generated that include timestamps and deficiency specifics.	FR	Nutrient sensor readings, alert logging system, timestamps	Nutrient sensors are operational and accessible	Medium
The system shall compare optimal pH soil readings against detected pH levels.	D4 N-277	AC: Alerts generated within 10 seconds of out-of-range reading.	FR	pH sensor readings, reference optimal pH values, timestamps	pH sensors are calibrated and functional	High
The system shall trigger automatic watering when moisture is below threshold.	D4 N-274	AC: The system correctly activates/deactivates the automatic watering system at the appropriate time.	FR	Soil moisture sensor readings, threshold definitions, irrigation controller access	Soil moisture sensors are installed and operational	Highest
The system shall compare soil moisture readings against defined thresholds.	D4 N-273	AC: The system correctly and accurately detects low moisture levels.	FR	Soil moisture sensor readings, threshold	Thresholds for soil moisture are defined	High

				values, timestamps	and accessible	
The system shall timestamp and include outcome for each log entry.	D4 N-270	AC: Control logic logs are human-readable and retrievable chronologically.	FR	Action logs, timestamps, log templates, outcome data	Control logic can timestamp and log actions accurately	High
The system shall record every automated action executed by control logic.	D4 N-269	AC: Each automated action generates a log entry to store within 1 second.	FR	Control logic events, system log interface, timestamps	Automated control logic executes correctly	High
The system shall continuously compare sensor readings against expected readings.	D4 N-263	AC: When an anomaly occurs, it is successfully detected and logged.	FR	Sensor readings, expected reference values, timestamps	Expected sensor readings are defined and available	High
The system shall flag abnormal readings and generate alert messages.	D4 N-264	AC: When a value exceeds the expected range, an alert is issued within 2 seconds.	FR	Sensor readings, alert definitions, logging system	Alert thresholds are defined and accessible	High
The system shall log anomalies in storage for later review.	D4 N-265	AC: All anomaly sensor readings are timestamped with the sensor identified.	FR	Anomaly detection output, logging system, storage interface	Logging system has sufficient capacity	Medium
The system shall verify calibration accuracy against known reference values.	D4 N-260	AC: Sensor readings after the sensors are calibrated deviate less than ¬±5%.	FR	Calibration reference values, sensor readings, measurement tools	Calibration reference values are accurate	High
The system shall provide a calibration routine	D4 N-259	AC: Calibration is completed automatically upon startup	FR	Sensor IDs, calibration routines,	Sensors can be calibrated	Medium

for each sensor connected.		without user input.		software interface	automatically	
The system shall display a clear privacy policy describing all data collected and what it is used for.	D4 N-255	AC: Privacy policy must be reachable from the main interface screen.	FR	Privacy policy document, display system, UI access	Privacy policy is complete and visible	Medium
The system shall require consent from the user before collecting any data.	D4 N-256	AC: User consent must be explicitly recorded before sensors activate	FR	User input, consent logging, UI system	User consent is recorded before data collection	High
The system shall read soil moisture values from moisture sensors at specified intervals.	D4 N-251	AC: Soil moisture readings are collected every 10 minutes.	FR	Soil moisture sensor readings, timestamps, sensor IDs	Soil moisture sensors are operational and calibrated	Medium
The system shall store and timestamp soil moisture readings in local storage for later use	D4 N-252	AC: Stored readings are retrievable within 2 seconds.	FR	Local storage access, timestamps, sensor readings	Local storage is available and reliable	Medium
The system shall calibrate light sensors to ensure accuracy before readings are recorded.	D4 N-246	AC: Calibration occurs automatically at startup and automatically once per day.	FR	Light sensor calibration data, sensor IDs, measurement tools	Light sensors are functional and accessible	Medium
The system shall read light intensity from light sensors at specified intervals.	D4 N-245	AC: Light sensor readings are collected every 5 minutes.	FR	Light sensor readings, timestamps, sensor IDs	Light sensors are calibrated	Medium
The system shall store and timestamp light readings in local storage for later use.	D4 N-247	AC: Light sensor readings are timestamped and stored locally within 30 seconds of a sensor data	FR	Local storage access, timestamps, light sensor readings	Local storage is available and reliable	Medium

		collection interval.				
The system shall collect soil pH readings at specified intervals.	D4 N-240	AC: Soil pH readings are updated every 30 minutes.	FR	pH sensor readings, timestamps, sensor IDs	pH sensors are functional and calibrated	Medium
The system shall collect soil nutrient readings at specified intervals.	D4 N-239	AC: Soil nutrient readings are updated every 30 minutes.	FR	Soil nutrient sensor readings, timestamps, sensor IDs	Soil nutrient sensors are functional and calibrated	Medium
The system shall store and timestamp soil nutrient and pH readings in local storage for later use.	D4 N-241	AC: The system stores all timestamped sensor readings within 1 minute of data collection intervals.	FR	Local storage access, timestamps, pH and nutrient sensor readings	Local storage is available and reliable	Medium
The system shall store and timestamp temperature and humidity readings in local storage for later use.	D4 N-235	AC: The system stores all timestamped sensor readings within 1 minute of data collection intervals.	FR	Temperature and humidity sensor readings, timestamps, sensor IDs	Temperature and humidity sensors are functional and calibrated	Medium
The system shall read humidity from humidity sensors at specified intervals.	D4 N-234	AC: Humidity readings are captured every 5 minutes.	FR	Humidity sensor readings, timestamps, sensor IDs	Humidity sensors are functional and calibrated	Medium
The system shall display real-time data on the main dashboard.	D4 N-320	AC: All major data panels such as environment, alerts, and growth update continuously.	FR	Dashboard UI, real-time sensor data, rendering engine	Dashboard interface is operational and updates in real time	Medium

The system shall push notifications to the user,Äôs device reading thresholds are exceeded	D4 N-319	AC: Generated alerts are pushed to all linked devices and stored in history log.	FR	Device notification system, thresholds, user account	User devices can receive notifications	Medium
The system shall generate alerts when environmental readings exceed predefined thresholds.	D4 N-317	AC: Alerts trigger when thresholds are exceeded or not met.	FR	Environmental sensor readings, threshold definitions, alert system	Alert thresholds are correctly defined	Medium
The system shall display color-coded alerts to indicate plant or environment status.	D4 N-315	AC: Alerts correctly correspond to defined color conditions	FR	Alert system interface, color coding parameters, UI	Alert system is operational	Medium
The system shall generate and display graphical growth trends using plant data	D4 N-313	AC: Users are able to view plant growth trends within a selected time frame.	FR	Plant growth data, dashboard rendering engine	Plant growth data is accurate and accessible	Medium
The system shall store and apply user's notification preferences across sessions.	D4 N-311	AC: Notification changes persist after logging out and back in.	FR	User account, session storage	User account and preferences are stored	Medium
The system shall allow users to enable and disable notifications for alerts from the system.	D4 N-310	AC: Users can successfully toggle notification types on/off for each type of notification.	FR	User account, notification settings interface	Notification settings interface is functional	Medium
The system shall allow users to navigate between all menus using the main navigation menu.	D4 N-308	AC: Users can successfully navigate to all major sections of the app.	FR	Navigation system interface, menu hierarchy	Navigation menu is properly configured	Medium

The design prototype shall include all core UI components with navigation flow.	D4 N-307	AC: Each important data collection sensor has at least one corresponding screen in the prototype.	FR	UI prototype, component library, navigation flow	UI prototype reflects final design	Medium
The system shall transmit validated sensor readings to the Raspberry Pi every 10 seconds using a secure communication protocol	D4 N-304	AC: Sensor readings arrive at the Raspberry Pi within the 10-second interval.	FR	Sensor readings, secure transmission protocol, Raspberry Pi interface	Secure transmission protocol is available and functional	Medium
The system shall define minimum and maximum thresholds for each sensor type (e.g., soil moisture, pH) to ensure valid readings.	D4 N-302	AC: Thresholds persist after reboot and the system successfully discards values outside of valid reading range.	FR	Sensor readings, min/max thresholds, validation routine	Sensor thresholds are defined and validated	Medium
The schema must include attributes for sensor ID, and lighting values.	D4 N-301	AC: The schema accepts storage of sensor type and the light values collected from the light sensor.	FR	Database schema, sensor ID, lighting attributes	Database schema includes sensor ID and lighting attributes	Medium
The schema must include attributes for sensor ID, and pH values.	D4 N-300	AC: The schema for soil nutrient values supports storage of sensor type and pH values collected.	FR	Database schema, sensor ID, pH attributes	Database schema includes sensor ID and pH attributes	Medium
The schema must include attributes for sensor ID, and moisture percentages.	D4 N-299	AC: Fields for soil moisture in the schema align with sensor data.	FR	Database schema, sensor ID, moisture attributes	Database schema includes sensor ID and moisture attributes	Medium
The system must record all incoming sensor data along	D4 N-297	AC: Each sensor reading is stored and includes	FR	Sensor readings,	Sensor readings are	Medium

with their timestamps and sensor type.		timestamps and sensor type.		timestamps, sensor type	timestamped and correctly recorded	
The system must organize control action data in a structured format for analytics or use in UI.	D4 N-294	AC: Logs are queryable by their attributes such as description or timestamp.	FR	Control action logs, database storage, analytics interface	Control action logs are structured and accessible	Medium
The system must store each automated or manual control action along with its timestamp, type, and description.	D4 N-293	AC: Each stored action includes a timestamp, type, and description in the database.	FR	Control action logs, timestamps, type, description	Control action logs include all required details	Medium
The system must flag or reject data points that are missing, or outside threshold values.	D4 N-290	AC: Diagnostic reports are available and flagged data is logged in them.	FR	Sensor readings, validation rules, storage system	Data validation rules are defined and applied	Medium
The system must check each section of incoming sensor data against defined acceptable ranges before storage.	D4 N-289	AC: Incoming data is rejected and logged the moment it is received to the database.	FR	Incoming sensor data, defined acceptable ranges, storage system	Sensor readings fall within defined acceptable ranges	Low
The model must export measurements and dimensions in standard engineering units for reference in material selection.	D4 N-287	AC: Exported measurements are compatible with material sourcing documents	FR	Engineering unit standards, CAD software	Model measurements are accurate and in standard engineering units	Medium
The system must generate a to-scale model representing all hardware components and their arrangement.	D4 N-286	AC: Model accurately reflects component dimensions within ¬±1 mm.	FR	Hardware layout, CAD model software, measurement data	Hardware model accurately reflects component dimensions	Medium

Output must be legible and printable in A4 format.	D4 N-232	AC: Output is legible and formatted to fit on A4 paper.	NFR	Printing system, PDF generator, A4 page format	Printed outputs are legible and formatted for A4	Medium
Sensor accuracy must be ¬±1.0¬∞C for temperature readings.	D4 N-236	AC: Temperature sensor readings are accurate within ±1.0°C.	NFR	Temperature sensor calibration data, sensor readings, measurement tools	Temperature sensors are calibrated and accurate	High
Sensor accuracy must be ¬±5% for humidity readings.	D4 N-237	AC: Humidity sensor readings are accurate within ±5%.	NFR	Humidity sensor calibration data, sensor readings, measurement tools	Humidity sensors are calibrated and accurate	Medium
Analysis computations shall complete within 3 seconds per plant dataset.	D4 N-334	AC: Analysis computations complete within 3 seconds per plant dataset.	NFR	Plant sensor datasets, analysis software, computation system	Analysis software is functional and datasets are complete	Medium
Analysis accuracy shall achieve at least 95% when validated against known nutrient test results.	D4 N-333	AC: Analysis achieves at least 95% accuracy against known nutrient test results.	NFR	Historical sensor data, analysis software, reference test results	Reference nutrient test results are accurate and available	Medium
The trained model shall achieve a minimum prediction accuracy of 90% during validation testing.	D4 N-330	AC: Trained model achieves a minimum prediction accuracy of 90% during validation testing.	NFR	Trained ML model, validation dataset, performance metrics	Trained ML model is properly trained and validated	Medium
All endpoints should use HTTPS and enforce encryption of data during transit.	D4 N-328	AC: All API endpoints use HTTPS and encrypt data during transit.	NFR	Secure API configuration, HTTPS certificates,	All endpoints support HTTPS and	Medium

				encryption keys	encryption is enforced	
API response times shall not exceed 1 second for 95% of requests.	D4 N-327	AC: API response times do not exceed 1 second for 95% of requests.	NFR	API requests, server performance logs, monitoring tools	API server is responsive and monitored	Medium
Storing machine learning predictions in the cloud should not take longer than 500 ms.	D4 N-324	AC: Storing machine learning predictions in the cloud completes within 500 ms.	NFR	Cloud database access, timestamps, ML prediction payloads	Cloud storage is available and writing is reliable	Medium
The alert log must be retained for >= 30 days.	D4 N-280	AC: Alert logs are retained for at least 30 days.	NFR	Alert log storage system, timestamps, database configuration	Alert log storage is available and properly configured	Medium
Retrieving logs of control logic actions should take <= 2 seconds.	D4 N-272	AC: Retrieving logs of control logic actions completes within 2 seconds.	NFR	Control logic logs, database query system, timestamps	Control logic logs are accessible and retrievable quickly	Medium
The automatic watering system should respond within 15 seconds of unexpected soil moisture levels.	D4 N-285	AC: Automatic watering system responds within 15 seconds of unexpected soil moisture levels.	NFR	Soil moisture sensor readings, irrigation controller interface, threshold definitions	Irrigation system responds reliably to sensor readings	Medium
The uptime of the lighting control system is > 99%	D4 N-284	AC: Lighting control system uptime is greater than 99%.	NFR	Lighting controller uptime logs, maintenance reports	Lighting control system is operational and	Medium

					maintained	
The lighting accuracy remains within ¬±5% of target value.	D4 N-283	AC: Lighting maintains accuracy within ±5% of target value.	NFR	Light sensor readings, target lighting values, timestamps	Light sensors and controls are accurate and functional	Medium
The accuracy of pH level detection >= 90%	D4 N-279	AC: pH level detection accuracy is at least 90%.	NFR	pH sensor calibration data, sensor readings, reference standards	pH sensors are calibrated and readings are accurate	Medium
The automatic watering system reliability should be >= 99%	D4 N-276	AC: Automatic watering system reliability is at least 99%.	NFR	Soil moisture readings, irrigation system logs, controller interface	Automatic watering system is reliable and tested	Medium
The system shall evaluate soil moisture levels against threshold in < 5 seconds.	D4 N-275	AC: Soil moisture evaluation against thresholds completes in under 5 seconds.	NFR	Soil moisture sensor readings, threshold definitions, timestamps	Soil moisture thresholds are defined and accessible	Medium
The storage capacity for action logs should store >= 30 days	D4 N-271	AC: Action log storage retains data for at least 30 days.	NFR	Control action log storage system, timestamps, database access	Action log storage system has sufficient capacity	Medium
Stored anomaly detection alerts are retained >= 30 days.	D4 N-268	AC: Stored anomaly detection alerts are retained for at least 30 days.	NFR	Anomaly detection storage system, timestamps,	Anomaly detection storage is available and reliable	Highest

				database interface		
The false positive rate for anomaly detection is <= 5%.	D4 N-267	AC: False positive rate for anomaly detection does not exceed 5%.	NFR	Anomaly detection output, reference validation datasets, statistical analysis	Anomaly detection system is validated and accurate	Medium
The latency for anomaly detection is <= 2 seconds.	D4 N-266	AC: Anomaly detection latency does not exceed 2 seconds.	NFR	Sensor readings, anomaly detection system logs, timestamps	Anomaly detection system is responsive	Medium
The system uptime during calibration > 99%	D4 N-262	AC: System uptime during calibration is greater than 99%.	NFR	Calibration logs, system uptime monitoring	Calibration system is operational and monitored	High
Calibration routine shall complete within 5 seconds per sensor.	D4 N-261	AC: Calibration routine completes within 5 seconds per sensor.	NFR	Sensor calibration routine, timestamps, software logs	Sensor calibration routines are functional and timely	High
All stored data must comply with IEEE privacy and security standards.	D4 N-258	AC: All stored data comply with IEEE privacy and security standards.	NFR	Stored sensor data, database security configuration, IEEE standard references	Stored data complies with IEEE privacy and security standards	High
The privacy policy message for a user must load in <2 seconds.	D4 N-257	AC: Privacy policy loads for the user in under 2 seconds.	NFR	Privacy policy content, display interface, system performance logs	Privacy policy content is accessible and optimized	High

The uptime for soil moisture monitoring is > 99%	D4 N-254	AC: Soil moisture monitoring uptime is greater than 99%.	NFR	Uptime logs, soil moisture monitoring system, performance reports	Soil moisture monitoring system is functional and maintained	High
The sensor readings have a latency of < 1 second.	D4 N-253	AC: Sensor readings have a latency of less than 1 second.	NFR	Sensor readings, timestamp logs, latency monitoring system	Sensor readings are timely and reliable	Highest
The calibration algorithm completes within 30 seconds.	D4 N-250	AC: Calibration algorithm completes within 30 seconds.	NFR	Calibration algorithm, execution logs, sensor interface	Calibration algorithm functions correctly within time limits	High
The light sensor has an uptime of > 99%	D4 N-249	AC: Light sensor uptime is greater than 99%.	NFR	Light sensor logs, uptime monitoring reports, sensor ID	Light sensor system is functional and uptime monitored	Highest
The light sensor has an accuracy of ¬±5%.	D4 N-248	AC: Light sensor accuracy is within ±5%.	NFR	Light sensor accuracy reports, calibration data, reference measurements	Light sensors are calibrated and accurate	High
The data integrity of sensor reading is ensured so that no data loss occurs over a 24 hour period	D4 N-244	AC: Sensor data integrity is maintained with no data loss over 24 hours.	NFR	Sensor reading logs, data integrity checks, 24-hour storage	Data storage system ensures integrity over 24 hours	High

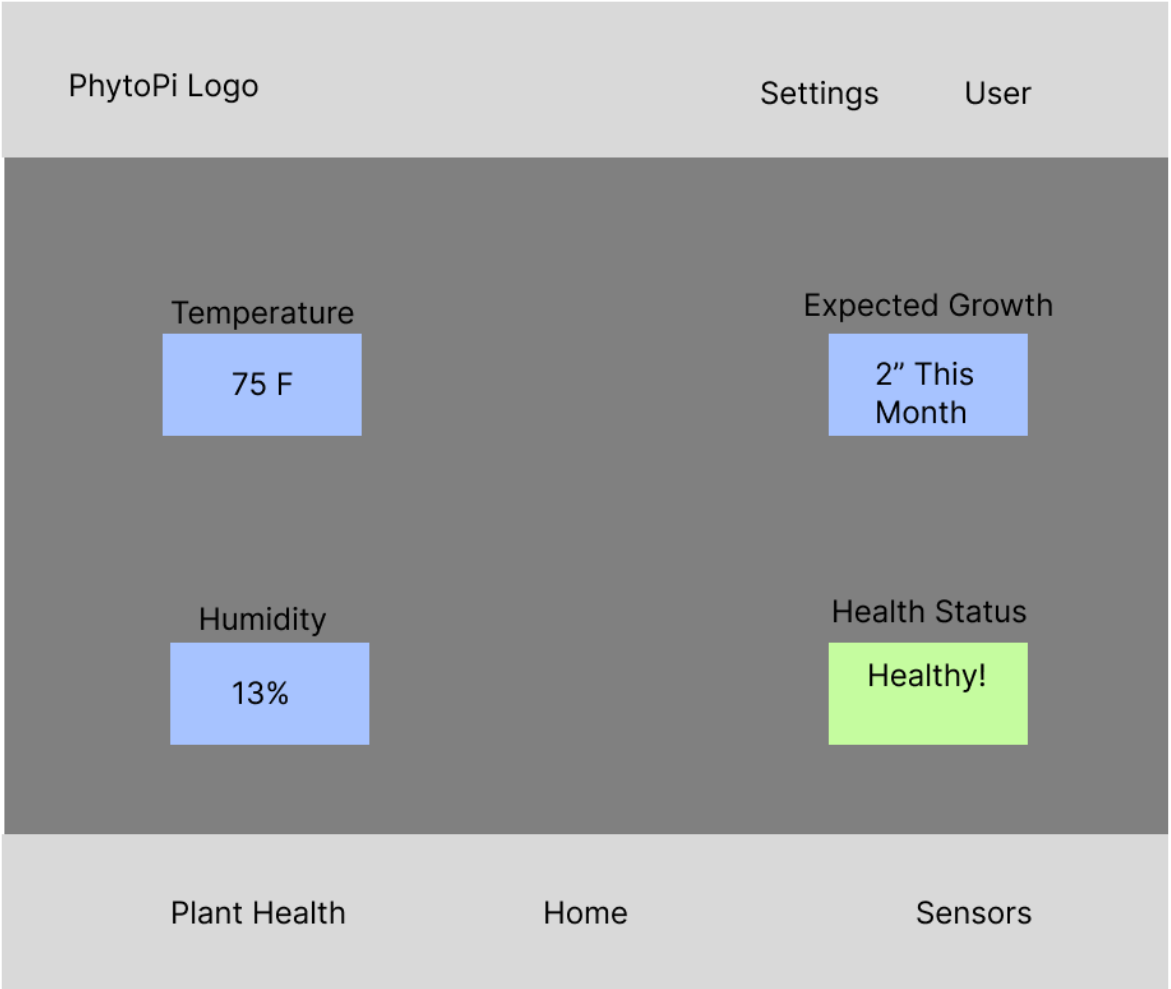
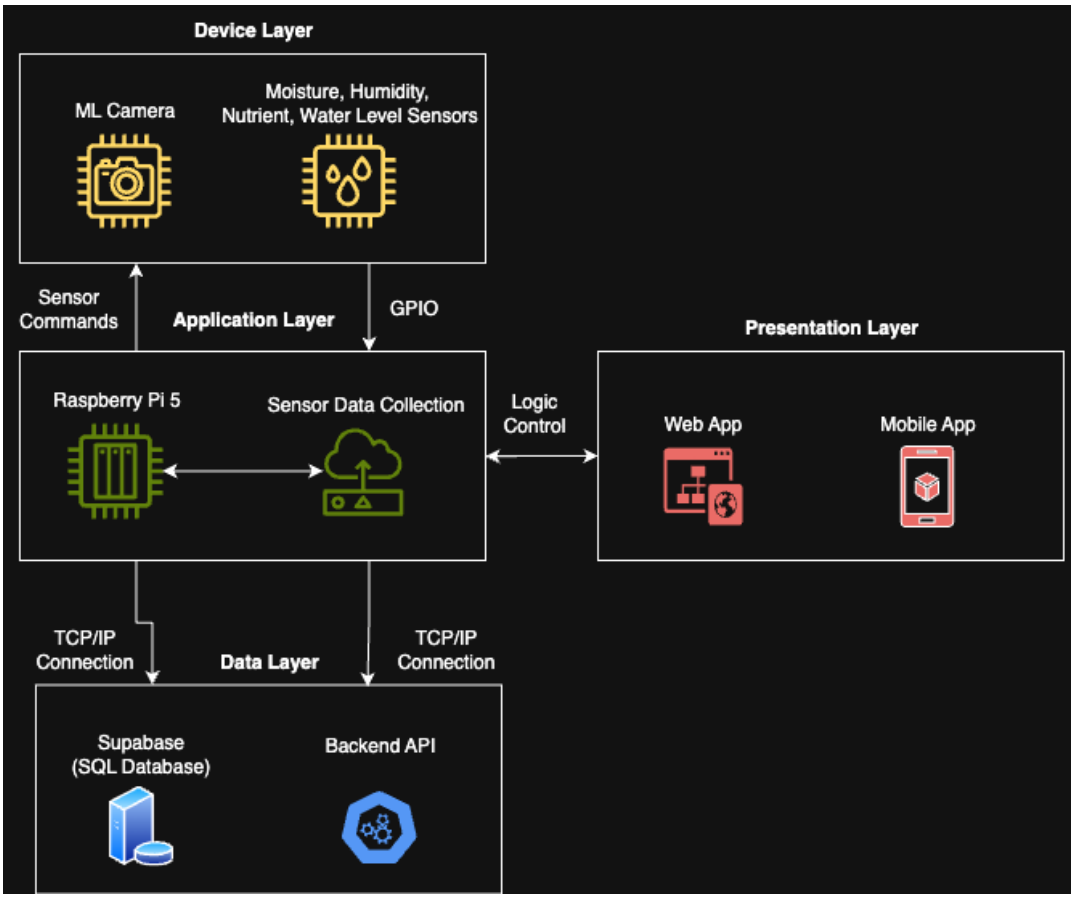
The pH sensor has an accuracy of ¬±0.2.	D4 N-243	AC: pH sensor accuracy is within ±0.2.	NFR	pH sensor calibration data, accuracy verification logs	pH sensors are calibrated and accurate	High
The soil nutrient sensor has an accuracy of ¬±5%	D4 N-242	AC: Soil nutrient sensor accuracy is within ±5%.	NFR	Soil nutrient sensor calibration data, accuracy verification logs	Soil nutrient sensors are calibrated and accurate	High
System uptime when collecting readings is >99%	D4 N-238	AC: System uptime during sensor readings is greater than 99%.	NFR	System uptime logs, sensor reading collection interface	Sensor collection system is reliable and maintaine d	High
Dashboard data refresh rate shall not exceed 1 second delay from actual sensor input.	D4 N-321	AC: Dashboard data refresh rate does not exceed 1 second delay from actual sensor input.	NFR	Dashboard UI, sensor input data, refresh rate monitoring	Dashboard UI is functional and refreshes correctly	High
Alerts shall be generated and delivered within 2 seconds of data detection.	D4 N-318	AC: Alerts are generated and delivered within 2 seconds of detection.	NFR	Alert generation system, detection timestamps, delivery logs	Alert generation system is responsive and accurate	High
Color changes shall occur within 500 milliseconds of alert status update.	D4 N-316	AC: Color-coded alert updates occur within 500 milliseconds.	NFR	UI color coding system, alert status updates, latency logs	UI color coding system is functional and consistent	High
Graphs shall render in under 5 seconds after time period selection	D4 N-314	AC: Graphs render in under 5 seconds after selecting a time period.	NFR	Graph rendering engine, dataset timestamps,	Graph rendering system is operationa l and	High

				performance logs	performant	
The notification settings shall have an error rate below 1% in usability testing.	D4 N-312	AC: Notification settings error rate is below 1% in usability testing.	NFR	Notification settings interface, usability testing logs, error reports	Notification settings system is accurate and reliable	High
Navigating between pages shall occur in under 1 second on a stable internet connection.	D4 N-309	AC: Navigating between pages occurs in under 1 second on a stable internet connection.	NFR	Navigation system interface, page load logs, performance monitoring	Navigation system is optimized and responsive	High
Data transmission latency shall not exceed 300 ms per packet.	D4 N-306	AC: Data transmission latency does not exceed 300 ms per packet.	NFR	Data transmission system, network packet logs, latency monitoring	Data transmission system is reliable and low-latency	High
The system shall maintain a minimum 99.5% successful transmission rate under stable network conditions.	D4 N-305	AC: System maintains a minimum 99.5% successful transmission rate under stable network conditions.	NFR	System transmission logs, success rate monitoring, stable network conditions	Network conditions are stable for expected transmission rates	High
Sensor threshold validation shall occur in under 100 ms per reading.	D4 N-303	AC: Sensor threshold validation occurs in under 100 ms per reading.	NFR	Sensor threshold validation system, reading timestamps, performance logs	Sensor threshold validation system is timely and functional	Medium
Database writing latency must not exceed 200ms	D4 N-298	AC: Database writing latency does not exceed 200 ms.	NFR	Database writing system, latency logs, performanc	Database writing system is responsive and monitored	Medium

				e monitoring		
Stored control logic actions must be retained in the database for >= 90 days	D4 N-296	AC: Stored control logic actions are retained in the database for at least 90 days.	NFR	Control logic action logs, database storage, retention policy	Control logic logs are retained per retention policy	Medium
Storing a received control logic action must occur in <= 500ms	D4 N-295	AC: Storing a received control logic action completes within 500 ms.	NFR	Control logic logging system, timestamps, storage latency monitoring	Control logic logging system is timely and reliable	Medium
Data validation accuracy >= 99%	D4 N-292	AC: Data validation achieves at least 99% accuracy.	NFR	Data validation system, accuracy reports, verification logs	Data validation system is accurate and verified	Medium
Data validation process must complete in under 100ms per reading.	D4 N-291	AC: Data validation process completes in under 100 ms per reading.	NFR	Data validation process logs, timestamps, performance monitoring	Data validation process is timely and performant	Medium
Model file size must not exceed 100 MB for sharing purposes.	D4 N-288	AC: Model file size does not exceed 100 MB for sharing purposes.	NFR	ML model file, file size logs, sharing records	ML model file is within size limits and shareable	Low

Wireframe for Dashboard

Main Dashboard



System Design